

System Monitoring — Master Technical Handbook

Complete End-to-End Deep Architectural & Technical Analysis (30 Sections)

Go / Gin Microservices

Python FastAPI & YOLOv8

React + TypeScript

PostgreSQL & Supabase

WebSockets & WebRTC

Interview Q&A Guide

Project: AI-Powered Secure Coding Assessment & Proctoring Platform

Codename: CodeShield / System Monitoring

Analysis Date: September 2026

Codebase Location: `c:\Users\Hp\OneDrive\Documents\GitHub\System-monitoring-`

TABLE OF CONTENTS

1. [Complete Project Inventory](#)
2. [Complete System Architecture](#)
3. [Frontend — Complete Technical Breakdown](#)
4. [Backend — Complete Internal Architecture](#)
5. [Database — Complete Deep Dive](#)
6. [Authentication & Authorization](#)
7. [Secure Coding Assessment System](#)
8. [Anti-Cheating / Security Monitoring](#)
9. [AI / Computer Vision — Extremely Deep Explanation](#)
10. [Real-Time Communication](#)
11. [Malpractice Detection Pipeline](#)
12. [Email Notification System](#)
13. [Supabase](#)
14. [Performance Engineering](#)
15. [Security Engineering](#)
16. [Deployment & DevOps](#)

17. [Error Handling](#)
 18. [Complete End-to-End User Flows](#)
 19. [File-by-File Technical Mapping](#)
 20. [Dependency Analysis](#)
 21. [Data Flow & Data Contracts](#)
 22. [Technical Decision Analysis](#)
 23. [Implemented vs Proposed](#)
 24. [Technical Limitations](#)
 25. [Scalability](#)
 26. [Technical Jury / Interview Questions](#)
 27. ["Why Did You Build It This Way?" Questions](#)
 28. [Complete Terminology Glossary](#)
 29. [Final Architecture Diagram](#)
 30. [30-Second / 1-Minute / 3-Minute Explanations](#)
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1. COMPLETE PROJECT INVENTORY

1.1 High-Level Directory Structure

```
System-monitoring-/  
├─ frontend/           ← React + TypeScript + Vite SPA (Candidate & Admin UI)  
├─ backend-auth/       ← Go + Gin primary REST API (Auth, Compiler, Timer, Malpractice)  
├─ Anticheating/       ← Go + Gin secondary microservice (Session monitoring, heartbeat)  
├─ yolo-proctor/       ← Python + FastAPI + YOLOv8 WebSocket server (Object detection AI)  
├─ render.yaml         ← Render.com multi-service deployment manifest  
├─ vercel.json          ← Vercel frontend deployment configuration  
├─ start-dev.bat       ← Windows batch script to launch all 4 services  
├─ yolo8n.pt           ← YOLOv8 Nano pre-trained model weights (6.5 MB)  
├─ pyrightconfig.json  ← Python type-checking config  
└─ .gitignore          ← Git exclusion rules
```

1.2 The Four Microservices

#	Service	Language	Port	Purpose
1	backend-auth	Go (Gin)	8080	Primary API: Auth, Compiler, Problems, Timer, Malpractice, Email
2	Anticheating	Go (Gin)	8081	Secondary API: Session monitoring, heartbeat, offline detection
3	yolo-proctor	Python (FastAPI)	8082	AI: YOLOv8 object detection via WebSocket
4	frontend	TypeScript (React/Vite)	5173	SPA: Candidate exam UI, Admin dashboard, AI camera widget

1.3 Every Source File in `backend-auth/`

File	Lines	Responsibility
main.go	70	Application entry point; wires all dependencies via constructor injection
config/config.go	126	Loads all env vars into a typed <code>Config</code> struct; custom <code>.env</code> parser
config/database.go	201	PostgreSQL/SQLite init with triple fallback; auto-migration; seed data
models/models.go	383	All GORM models and DTOs: Admin, User, ExamSession, Problem, TestCase, Submission, Leaderboard, MalpracticeLog
models/timer.go	99	ExamTimer model with <code>GetRemainingSeconds()</code> calculation and <code>ToResponse()</code>
routes/routes.go	123	All API route registration with Gin; groups: auth, admin, exam, compiler, monitoring, leaderboard
middleware/auth.go	146	Three JWT middleware functions: <code>RequireAuth</code> (user), <code>RequireAdmin</code> , <code>RequireAnyAuth</code>
middleware/logging.go	42	Request logger middleware; CORS middleware with wildcard <code>*</code> origin
controllers/auth_controller.go	126	Handlers: Register, Login, Logout, AdminRegister, AdminLogin
controllers/compiler_controller.go	113	Handlers: RunCode, SubmitCode; integrates leaderboard updates
controllers/exam_controller.go	~70	Handlers: StartExam, GetStatus, EndExam
controllers/malpractice_controller.go	84	Handlers: LogMalpractice, GetUserMalpractices, GetAllMalpractices
controllers/timer_controller.go	126	Handlers: GetTimerStatus, VerifyExamPassword, ConfigTimer, StartTimer, PauseTimer, ResumeTimer, ExtendTimer, EndTimer
controllers/problem_controller.go	~70	Handlers: GetAllProblems, GetProblemByID, CreateProblem
controllers/leaderboard_controller.go	~40	Handler: GetLeaderboard
controllers/user_controller.go	~55	Handlers: GetProfile, UpdateProfile
services/auth_service.go	201	Business logic: Register, AuthenticateUser, RegisterAdmin, AuthenticateAdmin, ValidateToken, generateToken
services/compiler_service.go	502	Code execution engine: Go, Python, JavaScript/TS, Java, C++, Rust via <code>os/exec</code>

File	Lines	Responsibility
services/email_service.go	239	Resend API email sender; builds professional HTML alert emails
services/malpractice_service.go	99	Violation logging with auto-lookup of candidate info; async email dispatch
services/timer_service.go	198	Central exam timer: configure, start, pause, resume, extend, end, verify password
services/exam_service.go	128	Exam session lifecycle: start, get status, end, auto-expire
services/leaderboard_service.go	95	Leaderboard ranking and upsert logic
services/problem_service.go	115	Problem CRUD with test case management
services/user_service.go	64	Profile get and update (protects email/password from modification)
repositories/*.go	8 files	GORM database operation wrappers for each model
pkg/utls/utls.go	57	Standardized JSON response helpers: Success, Error, BadRequest, Unauthorized, Forbidden, NotFound, InternalError, Conflict

1.4 Every Source File in `frontend/src/`

File	Size	Responsibility
main.tsx	246B	React DOM entry; renders <code><App /></code> in StrictMode
App.tsx	2.5KB	Root component; BrowserRouter, providers (Auth→Monitoring→Exam), theme switching
api/client.ts	59KB	Massive unified API layer: real backend calls with localStorage offline fallbacks
contexts/AuthContext.tsx	3.2KB	Auth state: user profile, login/register/logout, token rehydration
contexts/ExamContext.tsx	8.2KB	Exam state: problems, code editor map, timer sync (1.5s polling), run/submit
contexts/MonitoringContext.tsx	10KB	Security monitoring state: confidence score, risk, violations, fullscreen, keyboard lock, copy/paste protection
hooks/useAntiCheating.ts	10.5KB	18+ browser event listeners: context menu, clipboard, keydown, fullscreen, visibility, blur/focus, drag, scroll, DevTools
hooks/useCopyPasteProtection.ts	7.9KB	Deep clipboard/selection/drag prevention with CSS <code>user-select:none</code> injection
routes/AppRoutes.tsx	2.7KB	React Router v7 route definitions; <code>AdminOnly</code> guard component
types/index.ts	3.8KB	All TypeScript interfaces: UserProfile, ProblemData, MonitoringEvent, CompilerResult, LeaderboardEntry, etc.
services/mockData.ts	12.9KB	Seed data: candidate cards, problems with starter code, monitoring events, leaderboard
components/monitoring/AICameraWidget.tsx	30KB	Core AI component: webcam stream, COCO-SSD in-browser detection, MediaPipe FaceMesh, YOLO WebSocket, canvas overlay
components/compiler/MonacoWrapper.tsx	4.4KB	Monaco Editor integration with language switching
components/compiler/ConsoleOutput.tsx	11.2KB	Console output panel with test case results
components/compiler/ProblemDescription.tsx	5.1KB	Problem statement display
components/dashboard/*.tsx	7 files	Admin components: AdminTimerControl, CreateProblemModal, LiveCandidateCard, CandidateDetailModal, StatCard, ActivityChart, AdminManagementPanel

File	Size	Responsibility
components/monitoring/*.tsx	6 files	Monitoring overlays: ExamPasswordGateModal, FullscreenOverlay, LockScreenOverlay, SecurityViolationModal, SuspiciousTimeline
pages/Auth/*.tsx	4 files	CandidateLogin, CandidateRegister, AdminLogin, AdminRegister
pages/Admin/AdminDashboard.tsx	13.7KB	Admin dashboard with timer control, live candidates, malpractice logs
pages/Candidate/*.tsx	5 files	CandidateDashboard, AssessmentsPage, ProblemsPage, ResultsPage, AssessmentReadinessPage
pages/Compiler/ExamPage.tsx	10KB	Main exam page: Monaco Editor + AI Camera + Timer + Problem panel
pages/Compiler/SandboxPage.tsx	7.7KB	Practice/sandbox coding environment

1.5 Every Environment Variable

Variable	Default	Where Used	Purpose
<code>SERVER_PORT</code> / <code>PORT</code>	<code>8080</code>	backend-auth	HTTP server listen port
<code>APP_ENV</code>	<code>development</code>	backend-auth	Controls Gin mode (release vs debug)
<code>DB_HOST</code>	<code>aws-1-ap-south-1.pooler.supabase.com</code>	backend-auth	PostgreSQL host (Supabase pooler)
<code>DB_PORT</code>	<code>5432</code>	backend-auth	PostgreSQL port
<code>DB_USER</code>	<code>postgres.wdtshwffstjzfc1wsqox</code>	backend-auth	Supabase DB user
<code>DB_PASSWORD</code>	<code>Kishores%402029</code>	backend-auth	DB password (URL-encoded @)
<code>DB_NAME</code>	<code>postgres</code>	backend-auth	Database name
<code>DB_SSLMODE</code>	<code>require</code>	backend-auth	TLS mode for Postgres connection
<code>JWT_SECRET</code>	<code>codeshield-user-secret-key- ...</code>	backend-auth	HMAC key for user JWT signing
<code>JWT_EXPIRY_HOURS</code>	<code>24</code>	backend-auth	User token lifetime
<code>ADMIN_JWT_SECRET</code>	<code>codeshield-admin-secret-key- ...</code>	backend-auth	HMAC key for admin JWT
<code>ADMIN_JWT_EXPIRY_HOURS</code>	<code>12</code>	backend-auth	Admin token lifetime
<code>COMPILER_TIMEOUT_SECONDS</code>	<code>5</code>	backend-auth	Max code execution time
<code>MAX_CODE_SIZE_KB</code>	<code>100</code>	backend-auth	Max code submission size (102,400 bytes)
<code>TEMP_DIR</code>	OS temp	backend-auth	Directory for code compilation temp files

Variable	Default	Where Used	Purpose
RESEND_API_KEY	(empty)	backend-auth	Resend email API key
MAIL_FROM	onboarding@resend.dev	backend-auth	Email sender address
VITE_API_BASE	auto-detect	frontend	Backend API URL
VITE_ANTICHEATING_API_BASE	auto-detect	frontend	Anticheating service URL
VITE_YOLO_WS_URL	auto-detect	frontend	YOLO WebSocket URL

2. COMPLETE SYSTEM ARCHITECTURE

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